VERY AMPLENESS IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The projective bundle and its twisting sheaf are compatible with base change. Constructions, Lemma \ref{constructions-lemma-relative-proj-base-change}, cited at proof lines 9257-9260. Identifies the base change of the projective bundle and of the twisting sheaf.
- The base change of an immersion is an immersion. Schemes, Lemma \ref{schemes-lemma-base-change-immersion}, cited at proof lines 9261-9263. Produces the immersion witnessing very ampleness after base change.
- The definition of a very ample invertible sheaf as a pullback of the twisting sheaf along an immersion into a projective bundle. Definition \ref{definition-very-ample}, cited at proof lines 9249-9254. Supplies the data that is transported along the base change.

Source reference. Original lemma 0B3F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13963–13963. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2451–2451. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01O3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2180–2180. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01JY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2031–2031. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $S' \to S$ be a morphism of schemes. Let $f' : X' \to S'$ be the base change of f and denote $\mathcal{L}'$ the pullback of $\mathcal{L}$ to X'. If $\mathcal{L}$ is f-very ample, then $\mathcal{L}'$ is f'-very ample.*

Proof. By Definition 01VM there exists there exist a quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$ and an immersion $i : X \to \mathbf{P}(\mathcal{E})$ over S such that $\mathcal{L} \cong i^*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1)$. The base change of $\mathbf{P}(\mathcal{E})$ to S' is the projective bundle associated to the pullback $\mathcal{E}'$ of $\mathcal{E}$ and the pullback of $\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1)$ is $\mathcal{O}_{\mathbf{P}(\mathcal{E}')}(1)$, see Constructions, Lemma 01O3. Finally, the base change of an immersion is an immersion (Schemes, Lemma 01JY). □